



Physiology of Metabolism

Part Two

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Learning Outcomes:

- Discuss the fate and metabolism of lipids.
- Describe the fate and metabolism of proteins.

Lipid Metabolism

Lipid Catabolism

Lipolysis

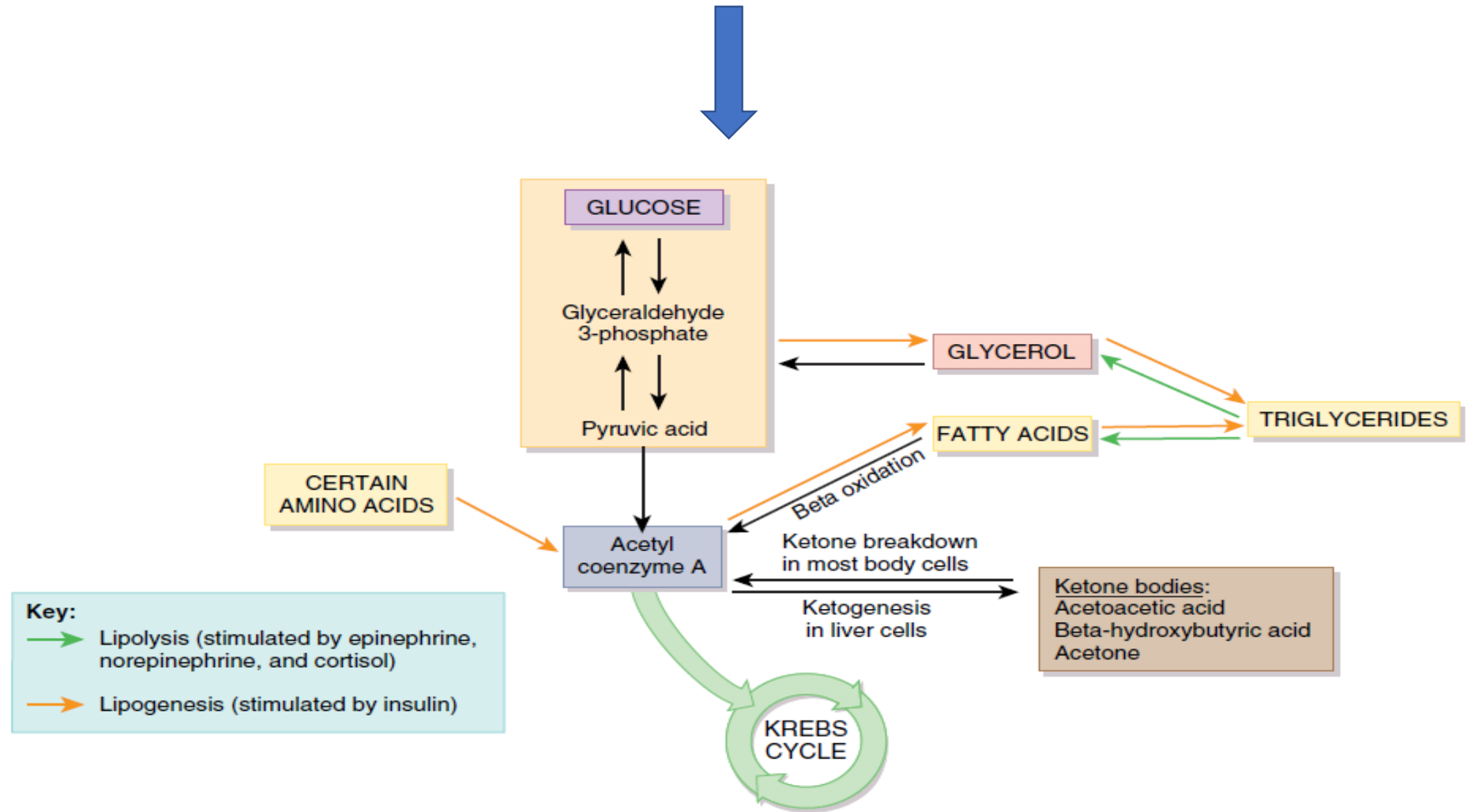
Lipid Anabolism

Lipogenesis

Lipid Catabolism: Lipolysis

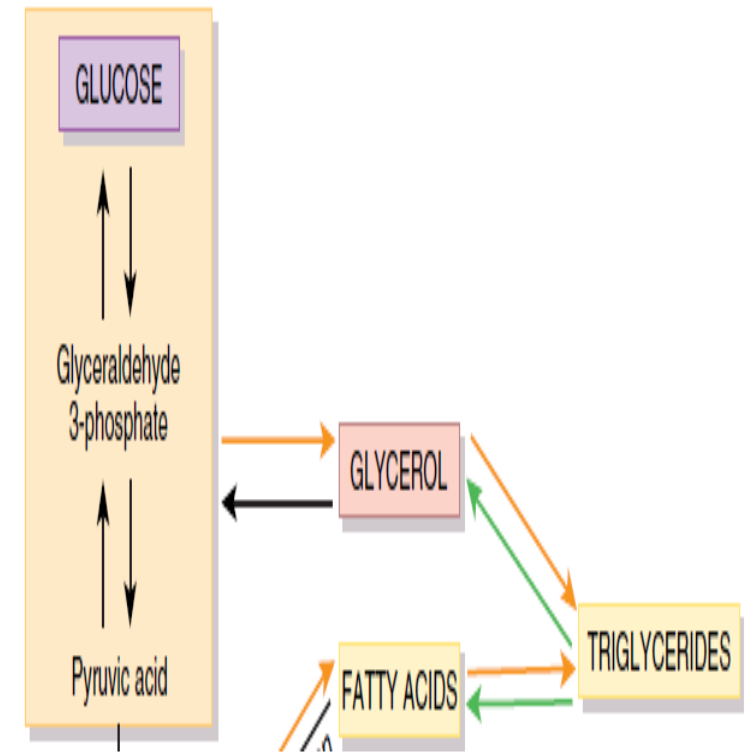
- In order for muscle, liver, and adipose tissue to oxidize the fatty acids derived from triglycerides to produce ATP, the triglycerides must first be split into glycerol and fatty acids, a process called **Lipolysis**.
- Lipolysis is catalyzed by enzymes called **lipases**.
- **Epinephrine** and **norepinephrine** enhance triglyceride breakdown into fatty acids and glycerol. These hormones are released when sympathetic tone increases, as occurs, for example, during exercise.
- Other lipolytic hormones include **cortisol**, thyroid hormones, and **insulinlike growth factors**. By contrast, **insulin inhibits lipolysis**.

The glycerol and fatty acids that result from lipolysis are catabolized via different pathways



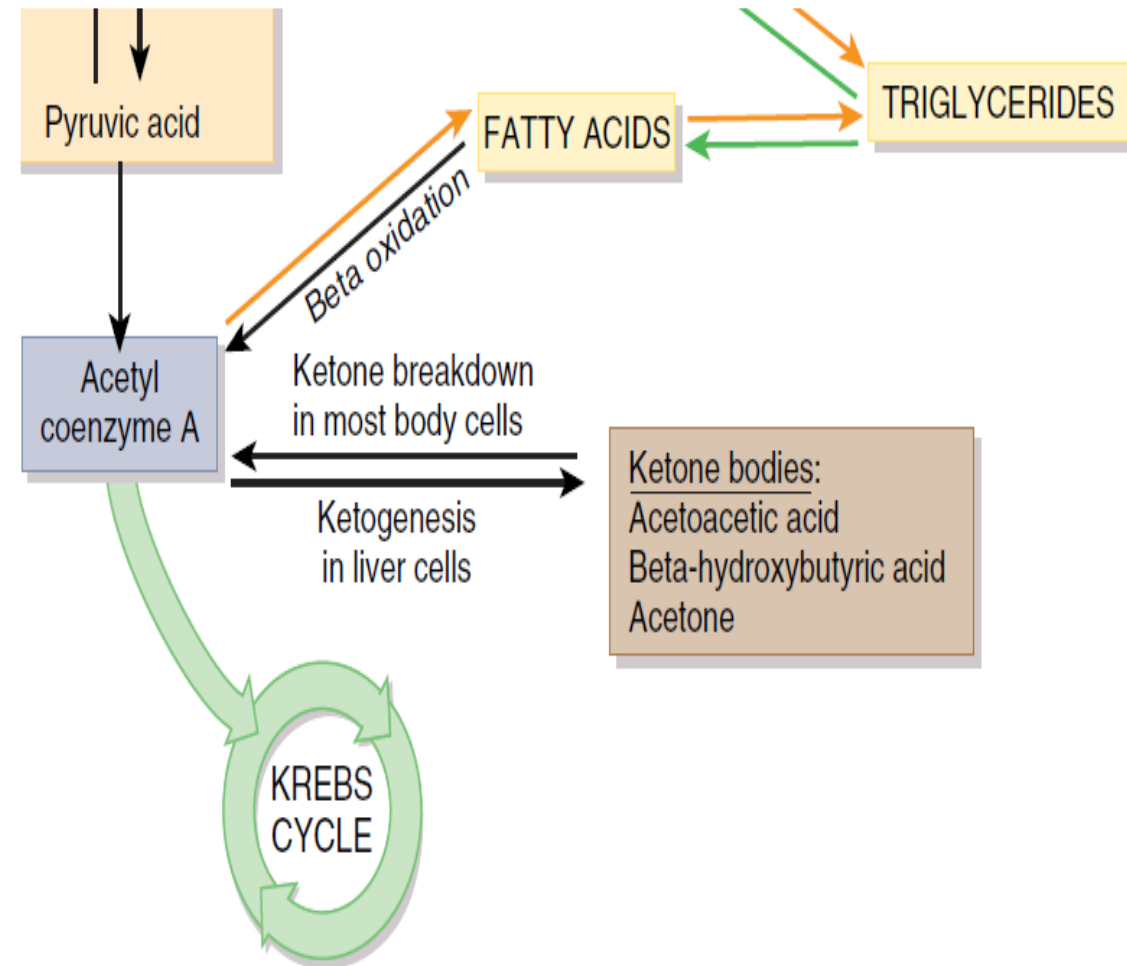
Glycerol Catabolism:

- If ATP supply in a cell is high  glycerol is metabolized into glucose, an example of gluconeogenesis.
- If ATP supply in a cell is low  glycerol enters the catabolic pathway to pyruvic acid.



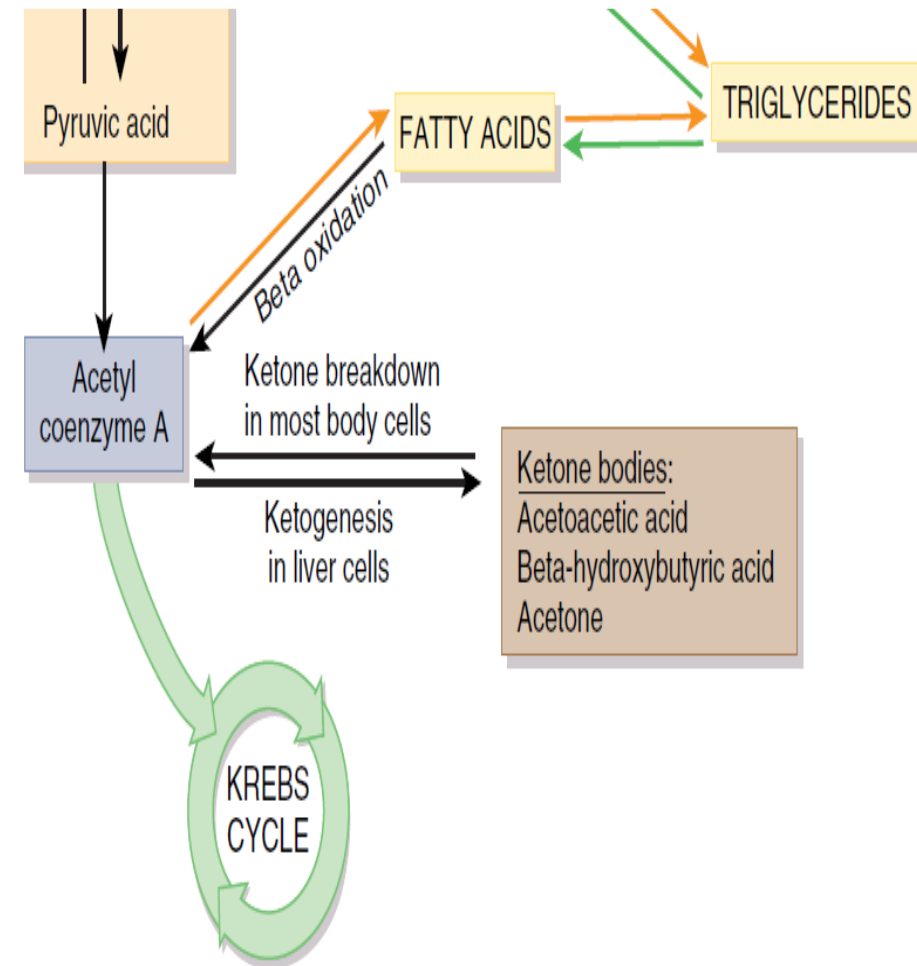
Fatty Acid Catabolism:

- Fatty acids are catabolized differently than glycerol and yield more ATP.
- fatty acid catabolism is called **beta oxidation**, that occurs in the matrix of mitochondria and leads to the formation of acetyl CoA and then acetyl CoA enters the Krebs cycle.
- As part of normal fatty acid catabolism, hepatocytes can take two acetyl CoA molecules at a time and condense them to form **acetoacetic acid**.



Fatty Acid Catabolism:

- Some acetoacetic acid is converted into beta-hydroxybutyric acid and acetone. The formation of these three substances, collectively known as **ketone bodies**, is called **Ketogenesis**.
- ketone bodies enter the bloodstream.
- Other cells take up acetoacetic acid and eventually enter the Krebs cycle.
- Heart muscle and the cortex (outer part) of the kidneys use acetoacetic acid in preference to glucose for generating ATP.

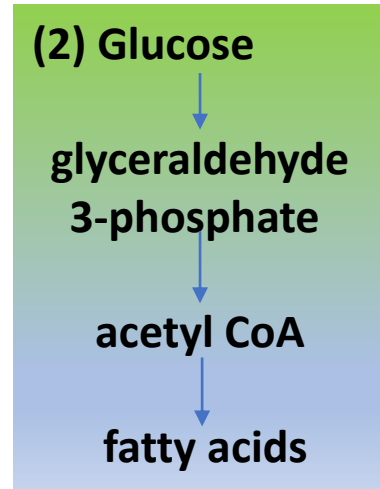
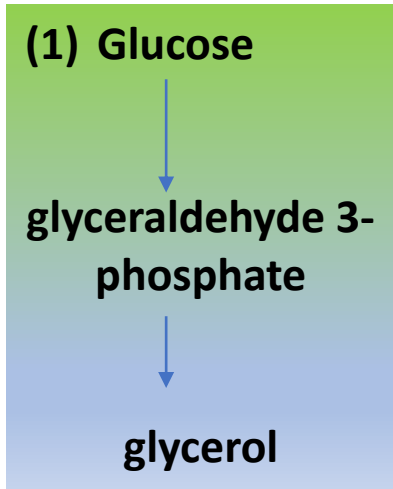


Lipid Anabolism: Lipogenesis

- Liver cells and adipose cells can synthesize lipids from glucose or amino acids through **lipogenesis**, which is stimulated by **insulin**.
- Lipogenesis occurs when individuals consume more calories than are needed to satisfy their ATP needs.
- Excess dietary carbohydrates, proteins, and fats all have the same fate  they are converted into triglycerides.

Lipogenesis:

- Certain amino acids converted to Fatty Acids then → triglycerides.
- glucose can be converted to either Glycerol or Fatty Acids then → become stored as **triglycerides**.



- or sometimes Glycerol or Fatty Acids undergo anabolic reactions to produce **other lipids** such as lipoproteins, phospholipids, and cholesterol.

CLINICAL CONNECTION | Ketosis

- The level of ketone bodies in the blood normally is very low because other tissues use them for ATP production.
- When the production of ketone bodies exceeds their uptake and use by body cells. EX: after a meal rich in triglycerides, or during fasting or starvation, because few carbohydrates are available for catabolism. Or in poorly controlled or untreated diabetes mellitus.

CLINICAL CONNECTION | Ketosis

- When the concentration of ketone bodies in the blood rises above normal a condition called **ketosis**—the ketone bodies, most of which are acids, must be buffered.
- If too many accumulate, they decrease the concentration of buffers and blood pH falls. Extreme or prolonged ketosis can lead to acidosis (**ketoacidosis**), an abnormally low blood pH.
- This in turn causes depression of the central nervous system, which can result in disorientation, coma, and even death if left untreated.
- When a diabetic becomes seriously insulin-deficient, one of the telltale signs is the sweet smell on the breath from the ketone body acetone.



Protein Metabolism

- During digestion, proteins are broken down into amino acids.
- Unlike carbohydrates and triglycerides, which are stored, proteins are not warehoused for future use.
- Amino acids are either oxidized to produce ATP or used to synthesize new proteins for body growth and repair.
- Excess dietary amino acids are not excreted in the urine or feces but instead are converted into glucose (gluconeogenesis) or triglycerides (lipogenesis)

Protein Catabolism

- A certain amount of protein catabolism occurs in the body each day, stimulated mainly by **cortisol** from the adrenal cortex.
- Some amino acids are converted into other amino acids, peptide bonds are re-formed, and new proteins are synthesized as part of the recycling process.

Protein Catabolism

- Hepatocytes convert some amino acids to fatty acids, ketone bodies, or glucose. Cells throughout the body oxidize a small amount of amino acids to generate ATP via the Krebs cycle.
- Before amino acids can enter the Krebs cycle, their amino group (NH_2) must first be removed, a process called **deamination**.
- Deamination occurs in hepatocytes and produces **ammonia (NH_3)**.
- The liver cells then convert the highly toxic ammonia to urea, a relatively harmless substance that is **excreted in the urine**.

Protein Anabolism

- Protein anabolism, the formation of peptide bonds between amino acids to produce new proteins, is carried out on the ribosomes of almost every cell in the body, directed by the cells' DNA and RNA.
- Insulinlike growth factors, thyroid hormones (T3 and T4), insulin, estrogen, and testosterone all stimulate protein synthesis.
- Because proteins are a main component of most cell structures, adequate dietary protein is especially essential during the growth years, during pregnancy, and when tissue has been damaged by disease or injury.

Protein Anabolism

- Of the 20 amino acids in the human body, 10 are **essential amino acids**: They must be present in the diet because they cannot be synthesized in the body in adequate amounts.
- It is essential to include them in your diet.
- Humans are **unable to synthesize** eight amino acids (isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine) and **synthesize two others (arginine and histidine) in inadequate amounts**, especially in childhood.

- A **complete protein** contains sufficient amounts of all essential amino acids.
- Beef, fish, poultry, eggs, and milk are examples of foods that contain complete proteins.
- An **incomplete protein** does not contain all essential amino acids. Examples of incomplete proteins are leafy green vegetables, legumes (beans and peas), and grains.
- Nonessential amino acids can be synthesized by body cells. They are formed by **transamination**  the transfer of an amino group from an amino acid to pyruvic acid or to an acid in the Krebs cycle.
- Once the appropriate essential and nonessential amino acids are present in cells, protein synthesis occurs rapidly.

Thank you